

UC2 Project Report

Dynamic Real Estate Optimization Using Forecasting and Scenario Simulation

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Context: UC-2: Sustainable Real Estate Asset Optimization (Morocco Case Study)

Project Overview

This project was designed as a decision-support system for long-term real estate planning under uncertainty. The objective was to help decision-makers evaluate how a public-sector real estate portfolio may evolve over a 15-year horizon and identify which strategies are most appropriate under different future conditions.

Large real estate portfolios often face fragmented visibility, uncertain future demand, rising operating costs, and difficult tradeoffs between maintaining, sharing, upgrading, or disposing of assets. A single forecast is not enough for these decisions, because planning depends on uncertainty in staffing, space utilization, and operating costs.

The goal of this project was therefore not just to predict future values, but to build an end-to-end workflow that supports **forecasting, stress testing, and decision-making**. The system focuses on three planning questions:

- How will building occupancy evolve over time?
- How will workforce demand change across regions?
- How will energy costs develop under different operating conditions?

These forecasts are translated into ranked strategic recommendations across five actions: **Build, Rent, Rehabilitate, Pool, and Dispose**.

So, the core practical question is:

Given uncertain future demand and cost, which real-estate strategy performs best under each policy preference and scenario and how confident should a decision-maker be in that recommendation?

And the answer might be :

Under this scenario, with these policy weights, this strategy has the highest modeled composite value.

Data Strategy

I identified several external sources that could have enriched the project, including GIS and satellite platforms such as ANCFCC, NASA, ArcGIS, Google Earth Engine, and OpenStreetMap, as well as rental-market platforms such as Airbnb, Mubawab, and Avito Maroc. I explored these sources as potential inputs for land-use patterns, urban growth, redevelopment activity, rental demand, and market pricing. However, I was not able to integrate them into the final pipeline because web

scraping and automated collection were blocked on those platforms, and much of the most relevant institutional data was not publicly accessible.

To keep the project end-to-end and methodologically consistent, I used a synthetic dataset calibrated to realistic public benchmarks. Calibration assumptions were informed by demographic, energy, and tariff benchmarks relevant to the Morocco context.

The synthetic dataset covered **57 buildings across 2005–2024** and included:

- building type, age, tenure, and surface area
- occupancy rate
- workforce headcount
- annual energy cost
- compliance and risk indicators

Methodology

The project was implemented as a multi-stage analytical pipeline:

- 1. Synthetic portfolio generation:** Created a building-level panel dataset with occupancy, workforce, energy cost, and risk indicators.
- 2. Occupancy forecasting:** Used SARIMA forecasting to model occupancy trajectories by building type, including uncertainty bands.
- 3. Workforce forecasting:** Used a cohort-based model to simulate retirement and hiring dynamics by region, with regression-based calibration.
- 4. Energy-cost forecasting:** Used OLS regression to estimate how energy cost responds to building size, age, climate, occupancy, and electricity price index.
- 5. Scenario simulation:** Combined occupancy, workforce, and energy signals in a scenario engine and ran Monte Carlo stress tests under four scenarios:
 - Optimistic
 - Austerity
 - Climate shock
 - Digitization
- 6. Strategy scoring and ranking** Converted scenario outcomes into ranked recommendations across cost efficiency, sustainability, and space flexibility.
- 7. Interactive dashboard** Presented outputs in a Streamlit dashboard with scenario selection, ranking views, sensitivity controls, and asset-level recommendations.

Key Results

The project produced three main outputs:

- long-range occupancy, workforce, and energy forecasts

- scenario-based stress-tested trajectories with uncertainty bands
- ranked strategy recommendations under each scenario

Key findings included:

- workforce and occupancy pressures changed the preferred portfolio strategy
- climate and energy shocks increased the rehabilitation-oriented decisions
- digitization and reduced space demand strengthened consolidation and disposal logic
- strategy rankings were sensitive to policy weights, showing that decision preferences meaningfully affect portfolio recommendations

Overall, the project showed that portfolio strategy should be tied to **future operating conditions**, not only current asset status.

A few examples from the scenario analysis:

- **Optimistic:** stronger workforce growth increased pressure for shared and flexible space strategies.
- **Austerity:** budget pressure favored consolidation and disposal of underused assets.
- **Climate shock:** rising energy costs increased the value of retrofitting older buildings.
- **Digitization:** reduced space need made pooling and disposal more attractive than expansion.

These results highlight the importance of scenario-based planning instead of relying on a single baseline forecast.

Streamlit Dashboard Demo

The final system includes a Streamlit dashboard that allows a user to:

- switch across future scenarios
- view long-range occupancy, workforce, and energy trajectories
- compare strategy rankings interactively
- adjust policy weights for cost efficiency, sustainability, and space flexibility
- inspect asset-level recommendations based on risk and utilization

Technical Takeaways

This project strengthened my applied skills in:

- time-series forecasting and stationarity analysis
- regression-based feature modeling
- Monte Carlo simulation for uncertainty analysis
- multi-criteria decision modeling
- dashboard design for communicating analytical outputs clearly

It also reinforced an important practical lesson: when production data is restricted, a carefully designed synthetic dataset can still support a realistic prototype of a full analytical system.

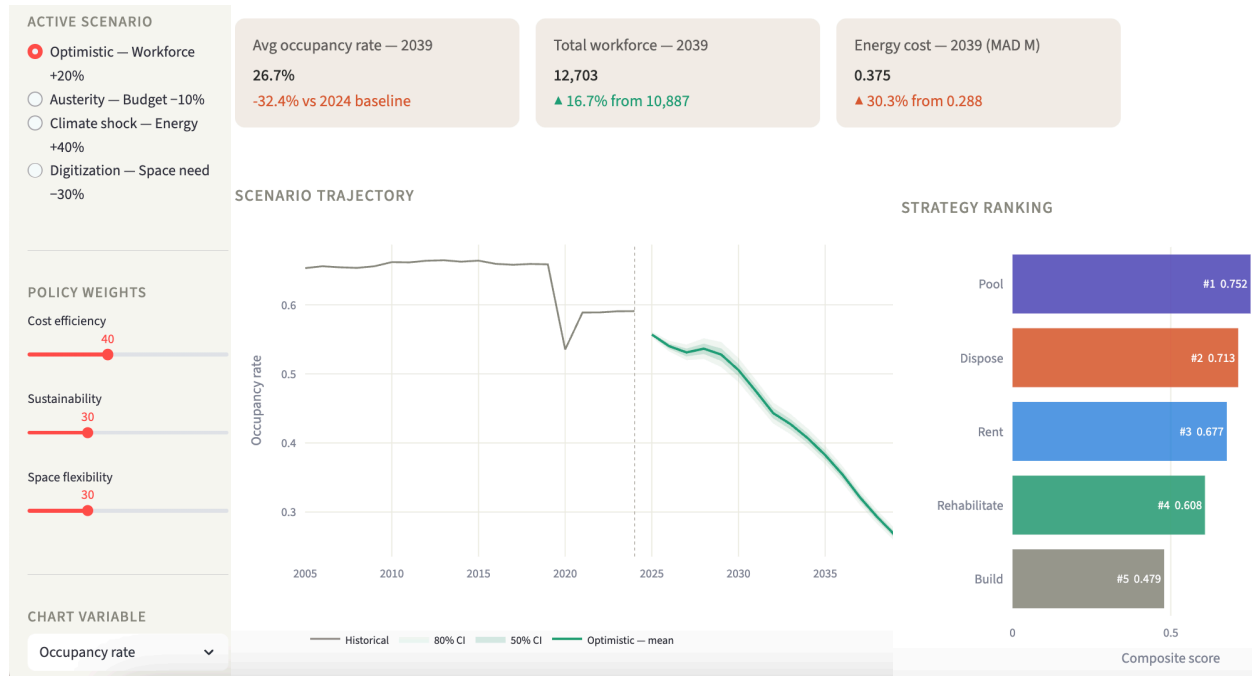


Figure 1: Streamlit dashboard for scenario-based real estate strategy analysis. The interface allows decision-makers to adjust policy weights, compare trajectories, and view ranked portfolio actions interactively.

Conclusion

The final result is a compact decision-support platform for long-horizon asset planning. Rather than treating forecasting as an isolated exercise, the project links data generation, modeling, simulation, and strategy ranking into one coherent workflow.